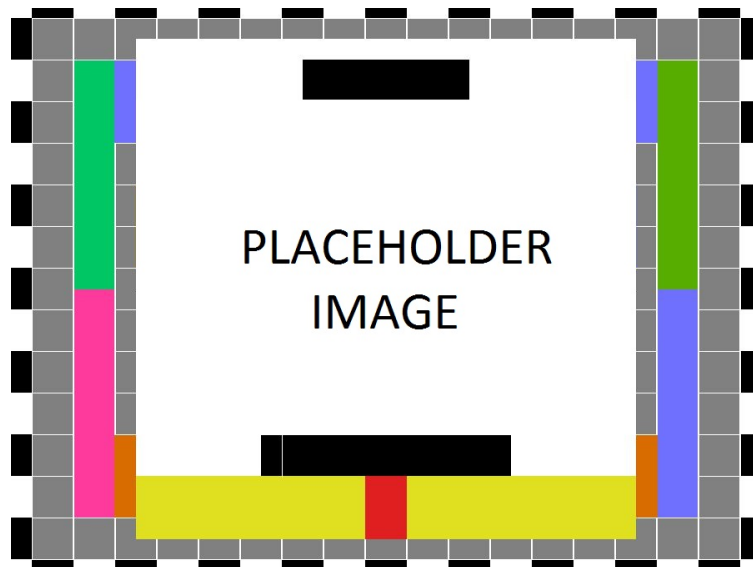


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**ARCHITECTURAL DESIGN SPECIFICATION
CSE 4316: SENIOR DESIGN I
FALL 2026**



**TEAM FULLSTACK
CIVICPULSE COMMUNITY SERVICE-REQUEST
DASHBOARD**

**ALAN TURING
GRACE HOPPER
JOHN VON NEUMANN**

ADA LOVELACE
CHARLES BABBAGE

REVISION HISTORY

Revision	Date	Author(s)	Description
0.1	11.02.2026	JVN	document creation
1.0	11.18.2026	AT, GH, CB	official release

CONTENTS

1	Introduction	5
2	Stakeholders & Architectural Concerns	5
3	Architecture Viewpoints	5
4	System Overview & Context View	5
5	Subsystem Definitions & Data Flow	5
6	Presentation Layer	5
6.1	Presentation.ResidentPortal	5
6.2	Presentation.StaffDashboard	6
7	Application Layer	6
7.1	Application.APIGateway	6
7.2	Application.RequestService	6
7.3	Application.NotificationService	6
8	Data Layer	6
8.1	Data.Datastore	6
8.2	Data.ObjectStore	6
8.3	Data.Integrations	7
9	Architecture Decisions & Rationale	7
9.1	AD-01: Three-tier with stateless API	7
9.2	AD-02: Delegated identity via OIDC	7
10	Requirements Traceability	7

1 INTRODUCTION

Architecture of CivicPulse Community Service-Request Dashboard as an ISO/IEC/IEEE 42010:2022 [1] description using IEEE 1016-2009 [2] viewpoints. Refines the SRS. Scope: the web frontend, backend API, and datastore, plus SSO/geocoding/SMTP integrations.

2 STAKEHOLDERS & ARCHITECTURAL CONCERNS

Stakeholder	Primary concerns
City (sponsor)	workflow fit, security, privacy, availability, cost
Residents	easy submission, accessibility, status visibility
Staff agents	efficient triage/assignment, role-appropriate access
City IT	SSO integration, observability, deployability

3 ARCHITECTURE VIEWPOINTS

Context (Section 4), Composition/Data Flow (Section 5), Logical/Functional (Sections 6–8), Deployment (Section 8). Style: a classic three-tier web application — SPA presentation, stateless API, relational + geospatial datastore — deployed as containers behind a reverse proxy. Interfaces #NN; subsystems [Layer] . [Name].

4 SYSTEM OVERVIEW & CONTEXT VIEW

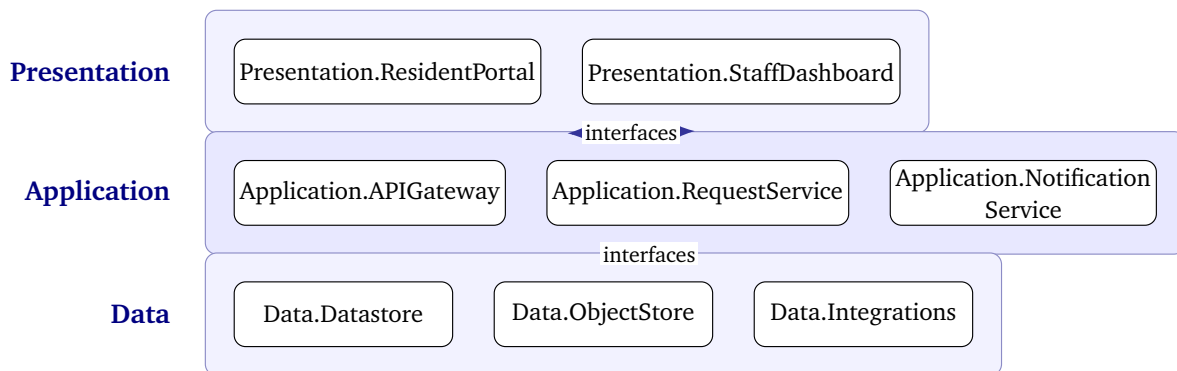


Figure 1: Three-tier web architecture

External actors: residents and staff (browsers), the city SSO (OIDC), the mapping API, and the SMTP relay. Presentation renders UI; Application enforces logic, auth, and integrations; Data persists data.

5 SUBSYSTEM DEFINITIONS & DATA FLOW

6 PRESENTATION LAYER

6.1 PRESENTATION.RESIDENTPORTAL

Responsibilities: public SPA for submit/track (FR-01, FR-02), map picker (EI-02), accessible/responsive (US-01).

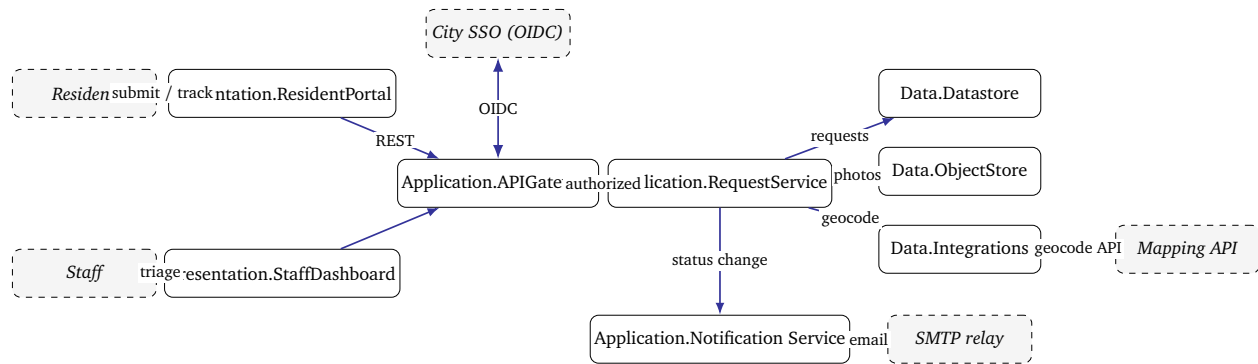


Figure 2: Browser → API → services → datastore/integrations

ID	Description	Inputs	Outputs
#01	Public REST calls	form data	request id, status

6.2 PRESENTATION.STAFFDASHBOARD

Responsibilities: authenticated SPA for triage/assignment/status (FR-03, FR-04); RBAC-aware UI (SE-01).

7 APPLICATION LAYER

7.1 APPLICATION.APIGATEWAY

Responsibilities: single REST entry, TLS termination, OIDC validation, RBAC enforcement (SE-01).

ID	Description	Inputs	Outputs
#02	Authenticated API	requests + token	authorized calls

7.2 APPLICATION.REQUESTSERVICE

Responsibilities: request lifecycle, triage, assignment, status history (FR-01–FR-04); duplicate detection (FR-05).

7.3 APPLICATION.NOTIFICATIONSERVICE

Responsibilities: send status emails via SMTP on status change (FR-04).

8 DATA LAYER

8.1 DATA.DATASTORE

Responsibilities: PostgreSQL + PostGIS for requests, users, audit, and geospatial queries (FR-05 radius). **Deployment:** managed DB in the cloud project.

8.2 DATA.OBJECTSTORE

Responsibilities: store request photos.

8.3 DATA.INTEGRATIONS

Responsibilities: adapters for the geocoding API and SMTP relay, isolating external dependencies behind stable interfaces.

9 ARCHITECTURE DECISIONS & RATIONALE

9.1 AD-01: THREE-TIER WITH STATELESS API

Context: multi-role web workflow, need for horizontal scale (PE-01, QA-01). **Decision:** stateless API behind a proxy with a shared datastore. **Alternatives:** server-rendered monolith (harder to scale/mobile). **Consequences:** clean SPA/API split; requires token-based auth. **Affects:** PE-01, SE-01, QA-01.

9.2 AD-02: DELEGATED IDENTITY VIA OIDC

Context: EI-01/SE-01 — staff use city credentials. **Decision:** delegate to the city IdP; store no staff passwords. **Alternatives:** local accounts (more risk, poor fit). **Consequences:** depends on SSO availability. **Affects:** EI-01, SE-01.

10 REQUIREMENTS TRACEABILITY

Req ID	Requirement (abbrev.)	Realizing subsystem(s)
FR-01/02	Submit/track	Presentation.ResidentPortal, Application.RequestService
FR-03	Triage/assign	Presentation.StaffDashboard, Application.RequestService
FR-04	Status + notify	Application.RequestService, Application.NotificationService
FR-05	Duplicate detection	Application.RequestService, Data.Datastore
SE-01	RBAC + TLS	Application.APIGateway
EI-01	City SSO	Application.APIGateway

REFERENCES

- [1] ISO/IEC/IEEE, “ISO/IEC/IEEE 42010:2022 — Software, systems and enterprise — Architecture description,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2022.
- [2] IEEE, “IEEE Std 1016-2009 — IEEE Standard for Information Technology — Systems Design — Software Design Descriptions,” Institute of Electrical and Electronics Engineers, New York, NY, USA, Tech. Rep., 2009.